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ASSIGNMENT No: 15

TITLE: Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement :Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives :To implement finally, throw, and throws for effective exception management. Theory The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Exercise

1. Create a Login program that throws an exception for invalid password and uses finally block.

```
import java.util.Scanner;
```

```
public class Login {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        String correctPassword = "admin123";
```

```
        try {
```

```
            System.out.print("Enter Password: ");
```

```

String password = sc.nextLine();

if (!password.equals(correctPassword))
    throw new Exception("Invalid Password!");

System.out.println("Login Successful.");

} catch (Exception e) {
    System.out.println("Error: " + e.getMessage());

} finally {
    System.out.println("Login process completed.");
    sc.close();
}
}
}

```

2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

```

import java.util.Scanner;

public class ATMVerification {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int correctPIN = 1234;

        try {
            System.out.print("Enter ATM PIN: ");
            int pin = sc.nextInt();

```

```
        if (pin != correctPIN)
            throw new Exception("Invalid PIN!");

        System.out.println("PIN Verified Successfully.");

    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());

    } finally {
        System.out.println("PIN verification process completed.");
        sc.close();
    }
}
```